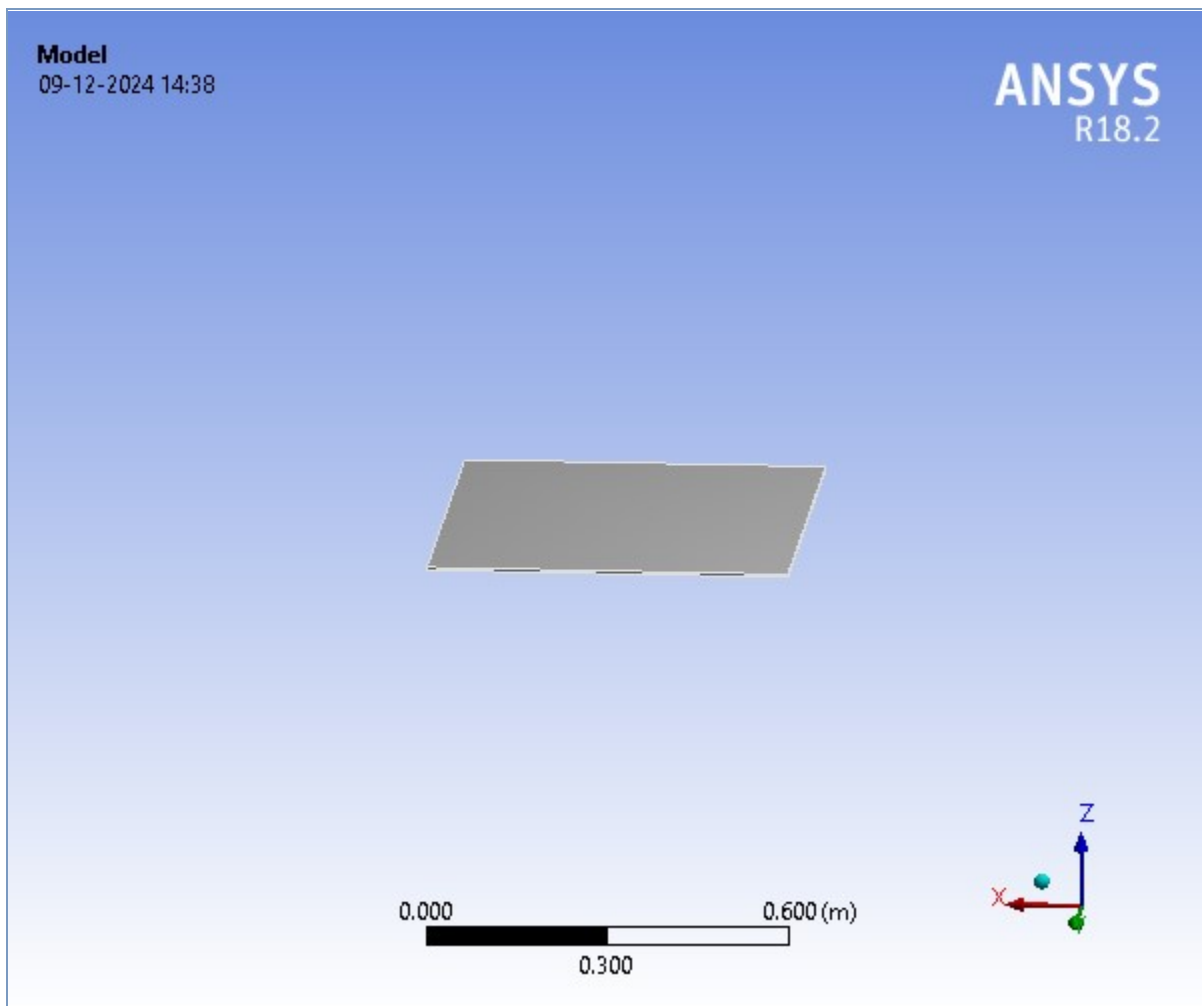




Project

First Saved	Friday, December 6, 2024
Last Saved	Monday, December 9, 2024
Product Version	18.2 Release
Save Project Before Solution	No
Save Project After Solution	No



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Report Not Finalized

Not all objects described below are in a finalized state. As a result, data may be incomplete, obsolete or in error. [View first state problem](#). To finalize this report, edit objects as needed and solve the analyses.

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (D4)

Geometry

TABLE 2

Model (D4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	D:\vfj 16nov\RAJ\wall&bullet_files\dp0\SYS-3\DM\SYS-3.agdb
Type	DesignModeler
Length Unit	Meters
Display Style	Body Color
Bounding Box	
Length X	0.6 m
Length Y	0.6 m
Length Z	3.9e-002 m
Properties	
Volume	1.8013e-003 m ³
Mass	14.052 kg
Scale Factor Value	1.
Statistics	
Bodies	4
Active Bodies	4
Nodes	8741
Elements	30572
Mesh Metric	None
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 3**Model (D4) > Geometry > Parts**

Object Name	Wall&bullet_____Final	Wall&bullet_____Final	Wall&bullet_____Final	Wall&bullet_____Final
State	Meshed			
Graphics Properties				
Visible	Yes			
Transparency	1			
Definition				
Suppressed	No			
Stiffness Behavior	Flexible			
Coordinate System	Default Coordinate System			

Reference Temperature	By Environment			
Reference Frame	Lagrangian			
Material				
Assignment	armer grade	Lead Core	Structural Steel	Brass
Bounding Box				
Length X	0.6 m	6.7951e-003 m	7.e-003 m	8.0764e-003 m
Length Y	0.6 m	6.7951e-003 m	7.e-003 m	8.0764e-003 m
Length Z	5.e-003 m	1.2124e-002 m	2.7e-002 m	3.2477e-002 m
Properties				
Volume	1.8e-003 m³	1.2139e-007 m³	8.3664e-007 m³	3.8022e-007 m³
Mass	14.04 kg	1.3717e-003 kg	7.0864e-003 kg	3.2204e-003 kg
Centroid X	3.3332e-019 m	-1.8975e-018 m	-9.9052e-008 m	3.8156e-008 m
Centroid Y	-7.0684e-018 m	-7.1614e-019 m	-9.2963e-007 m	-3.5549e-006 m
Centroid Z	-7.5e-003 m	-1.9117e-002 m	-3.2047e-002 m	-3.1559e-002 m
Moment of Inertia Ip1	0.42123 kg·m²	1.1573e-008 kg·m²	3.2049e-007 kg·m²	2.5992e-007 kg·m²
Moment of Inertia Ip2	0.42123 kg·m²	1.1573e-008 kg·m²	3.2048e-007 kg·m²	2.5992e-007 kg·m²
Moment of Inertia Ip3	0.8424 kg·m²	7.4381e-009 kg·m²	3.9112e-008 kg·m²	4.0475e-008 kg·m²
Statistics				
Nodes	7105	339	546	751
Elements	25123	1006	2222	2221
Mesh Metric	None			

Coordinate Systems

TABLE 4
Model (D4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Connections

TABLE 5
Model (D4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes

TABLE 6
Model (D4) > Connections > Contacts

Object Name	<i>Contacts</i>
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	2.1236e-003 m
Use Range	No
Face/Face	Yes
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	4
Active Connections	4

TABLE 7
Model (D4) > Connections > Contacts > Contact Regions

Object Name	Contact Region	Contact Region 2	Contact Region 3	Contact Region 4
State	Fully Defined			
Scope				
Scoping Method	Geometry Selection			
Contact	1 Face	2 Faces	4 Faces	7 Faces
Target	2 Faces		4 Faces	7 Faces
Contact Bodies	Wall&bullet_____Final			
Target Bodies	Wall&bullet_____Final			
Definition				
Type	Bonded			
Scope Mode	Automatic			
Behavior	Program Controlled			
Trim Contact	Program Controlled			
Trim Tolerance	2.1236e-003 m			
Maximum Offset	1.e-007 m			
Breakable	No			
Suppressed	No			

TABLE 8
Model (D4) > Connections > Body Interactions

Object Name	<i>Body Interactions</i>
State	Fully Defined
Advanced	
Contact Detection	Trajectory
Formulation	Penalty

Sliding Contact	Discrete Surface
Body Self Contact	Program Controlled
Element Self Contact	Program Controlled
Tolerance	0.2

TABLE 9**Model (D4) > Connections > Body Interactions > Body Interaction**

Object Name	<i>Body Interaction</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Frictionless
Suppressed	No

Mesh

TABLE 10**Model (D4) > Mesh**

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Body Color
Defaults	
Physics Preference	Explicit
Relevance	0
Element Order	Linear
Sizing	
Size Function	Adaptive
Relevance Center	Medium
Element Size	Default
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Slow
Initial Size Seed	Assembly
Span Angle Center	Coarse
Bounding Box Diagonal	0.849420 m
Minimum Edge Length	2.3073e-003 m
Quality	
Check Mesh Quality	Yes, Errors
Target Quality	Default (0.050000)
Smoothing	High
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No

Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	
Number of Retries	Default (4)
Rigid Body Behavior	Full Mesh
Mesh Morphing	Disabled
Triangle Surface Mesher	Program Controlled
Topology Checking	No
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	8741
Elements	30572

TABLE 11
Model (D4) > Mesh > Mesh Controls

Object Name	<i>Body Sizing</i>	<i>Face Meshing</i>	<i>Refinement</i>
State	Fully Defined	Ignored	Fully Defined
Scope			
Scoping Method	Geometry Selection	Geometry Selection	Geometry Selection
Geometry	3 Bodies	1 Face	1 Face
Definition			
Suppressed	No	No	No
Type	Element Size		
Element Size	1.5e-003 m		
Mapped Mesh		Yes	
Constrain Boundary		No	
Refinement			3
Advanced			
Defeature Size	Default		
Behavior	Soft		
Specified Sides		No Selection	
Specified Corners		No Selection	
Specified Ends		No Selection	

Explicit Dynamics (D5)

TABLE 12
Model (D4) > Analysis

Object Name	<i>Explicit Dynamics (D5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Explicit Dynamics
Solver Target	AUTODYN
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 13
Model (D4) > Explicit Dynamics (D5) > Initial Conditions

Object Name	<i>Initial Conditions</i>
-------------	---------------------------

State	Fully Defined
-------	---------------

TABLE 14
Model (D4) > Explicit Dynamics (D5) > Initial Conditions > Initial Condition

Object Name	Pre-Stress (None)	Velocity
State	Fully Defined	
Definition		
Pre-Stress Environment	None	
Pressure Initialization	From Deformed State	
Input Type		Velocity
Define By		Vector
Total		750. m/s
Direction		Defined
Suppressed		No
Scope		
Scoping Method		Geometry Selection
Geometry		3 Bodies

TABLE 15
Model (D4) > Explicit Dynamics (D5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Analysis Settings Preference	
Type	Custom
Step Controls	
Resume From Cycle	0
Maximum Number of Cycles	1e+07
End Time	1.e-004 s
Maximum Energy Error	0.1
Reference Energy Cycle	0
Initial Time Step	Program Controlled
Minimum Time Step	Program Controlled
Maximum Time Step	Program Controlled
Time Step Safety Factor	0.9
Characteristic Dimension	Diagonals
Automatic Mass Scaling	No
Solver Controls	
Solve Units	mm, mg, ms
Beam Solution Type	Bending
Beam Time Step Safety Factor	0.5
Hex Integration Type	Exact
Shell Sublayers	3
Shell Shear Correction Factor	0.8333
Shell BWC Warp Correction	Yes
Shell Thickness Update	Nodal
Tet Integration	Average Nodal Pressure
Shell Inertia Update	Recompute
Density Update	Program Controlled
Minimum Velocity	1.e-006 m s ⁻¹
Maximum Velocity	1.e+010 m s ⁻¹
Radius Cutoff	1.e-003
Minimum Strain Rate Cutoff	1.e-010
Euler Domain Controls	

Domain Size Definition	Program Controlled
Display Euler Domain	Yes
Scope	All Bodies
X Scale factor	1.2
Y Scale factor	1.2
Z Scale factor	1.2
Domain Resolution Definition	Total Cells
Total Cells	2.5e+05
Lower X Face	Flow Out
Lower Y Face	Flow Out
Lower Z Face	Flow Out
Upper X Face	Flow Out
Upper Y Face	Flow Out
Upper Z Face	Flow Out
Euler Tracking	By Body
Damping Controls	
Linear Artificial Viscosity	0.2
Quadratic Artificial Viscosity	1.
Linear Viscosity in Expansion	No
Artificial Viscosity For Shells	Yes
Hourglass Damping	AUTODYN Standard
Viscous Coefficient	0.1
Static Damping	0.
Erosion Controls	
On Geometric Strain Limit	Yes
Geometric Strain Limit	1.5
On Material Failure	No
On Minimum Element Time Step	No
Retain Inertia of Eroded Material	Yes
Output Controls	
Save Results on	Equally Spaced Points
Result Number Of Points	50
Save Restart Files on	Equally Spaced Points
Restart Number Of Points	5
Save Result Tracker Data on	Cycles
Tracker Cycles	1
Output Contact Forces	Off
Analysis Data Management	
Solver Files Directory	D:\vfj 16nov\RAJ\wall&bullet_files\dp0\SYS-3\MECH\
Scratch Solver Files Directory	

TABLE 16
Model (D4) > Explicit Dynamics (D5) > Loads

Object Name	<i>Fixed Support</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	4 Faces
Definition	
Type	Fixed Support
Suppressed	No

Solution (D6)

TABLE 17
Model (D4) > Explicit Dynamics (D5) > Solution

Object Name	<i>Solution (D6)</i>
State	Solved
Information	
Status	Done
Post Processing	
Beam Section Results	No

TABLE 18
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s
Display Points	All
Display Filter During Solve	Yes

TABLE 19
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Results

Object Name	Equivalent Stress	Equivalent Elastic Strain
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Equivalent (von-Mises) Stress	Equivalent Elastic Strain
By	Time	
Display Time	Last	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Integration Point Results		
Display Option	Averaged	
Average Across Bodies	No	
Results		
Minimum	0. Pa	0. m/m
Maximum	3.2337e+009 Pa	4.5307 m/m
Minimum Occurs On	Wall&bullet	Final
Maximum Occurs On	Wall&bullet	Final
Minimum Value Over Time		
Minimum	0. Pa	0. m/m
Maximum	0. Pa	0. m/m
Maximum Value Over Time		
Minimum	0. Pa	0. m/m
Maximum	9.2161e+009 Pa	4.5307 m/m
Information		
Time	1.0001e-004 s	
Set	51	
Cycle Number	8497	

FIGURE 1
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Equivalent Stress

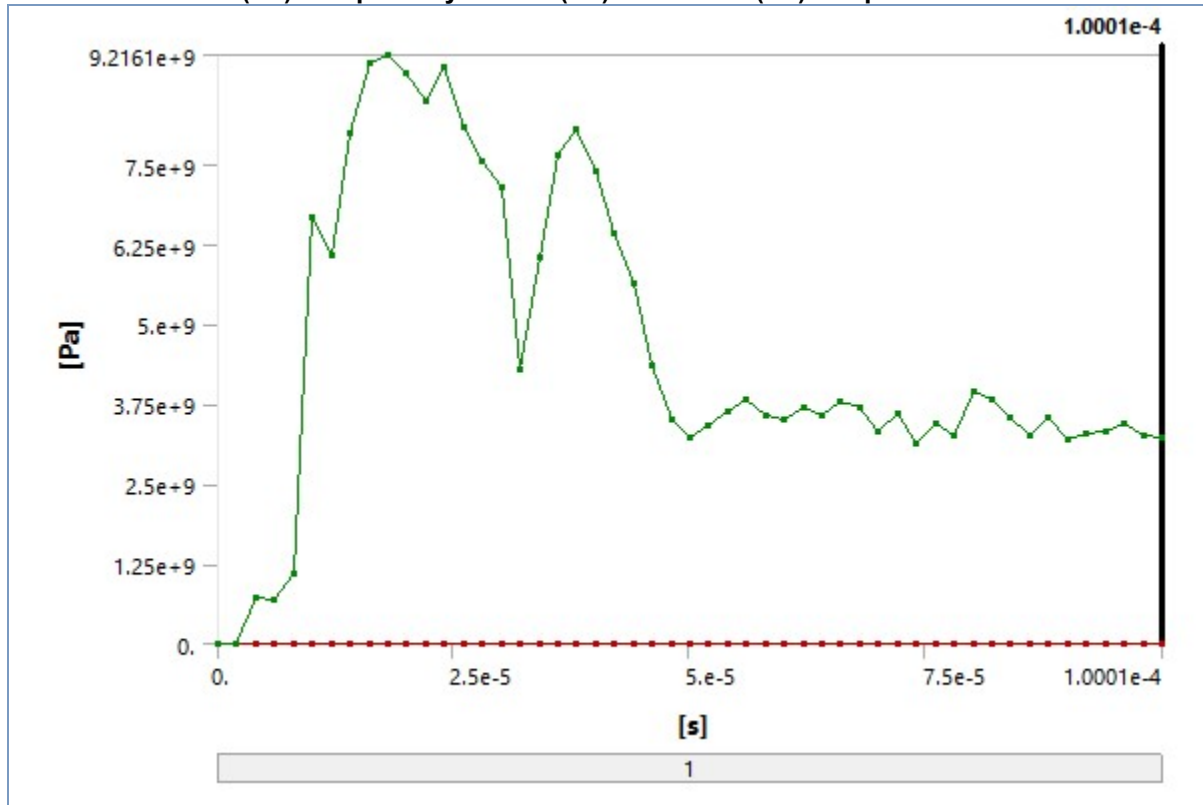


TABLE 20
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Equivalent Stress

Time [s]	Minimum [Pa]	Maximum [Pa]
1.1755e-038		0.
2.0034e-006		
4.0325e-006		7.2995e+008
6.0106e-006		6.9636e+008
8.0028e-006		1.1026e+009
1.0012e-005		6.6639e+009
1.2017e-005		6.076e+009
1.401e-005		7.9965e+009
1.6003e-005		9.0782e+009
1.8007e-005		9.2161e+009
2.0001e-005		8.9287e+009
2.2007e-005		8.4824e+009
2.4004e-005		9.0242e+009
2.6017e-005		8.0925e+009
2.8007e-005		7.5434e+009
3.0002e-005		7.1355e+009
3.2003e-005		4.3083e+009
3.4009e-005		6.0454e+009
3.6003e-005		7.6519e+009
3.8001e-005		8.0531e+009
4.0003e-005		7.4011e+009
4.2006e-005		6.4136e+009
4.4006e-005		5.639e+009

4.6008e-005	0.	4.357e+009
4.8009e-005		3.5013e+009
5.0004e-005		3.2312e+009
5.2005e-005		3.413e+009
5.4011e-005		3.6498e+009
5.601e-005		3.8392e+009
5.8e-005		3.5789e+009
6.0005e-005		3.5154e+009
6.2005e-005		3.6996e+009
6.4006e-005		3.5856e+009
6.6001e-005		3.7963e+009
6.8009e-005		3.6966e+009
7.0009e-005		3.3164e+009
7.2003e-005		3.6192e+009
7.4005e-005		3.144e+009
7.6002e-005		3.4468e+009
7.8004e-005		3.2711e+009
8.0009e-005		3.96e+009
8.2001e-005		3.8103e+009
8.4002e-005		3.5334e+009
8.6e-005		3.2514e+009
8.8002e-005		3.5473e+009
9.0007e-005		3.1999e+009
9.2002e-005		3.2801e+009
9.4002e-005		3.3187e+009
9.6e-005		3.4354e+009
9.8007e-005		3.2652e+009
1.0001e-004		3.2337e+009

FIGURE 2
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Equivalent Elastic Strain

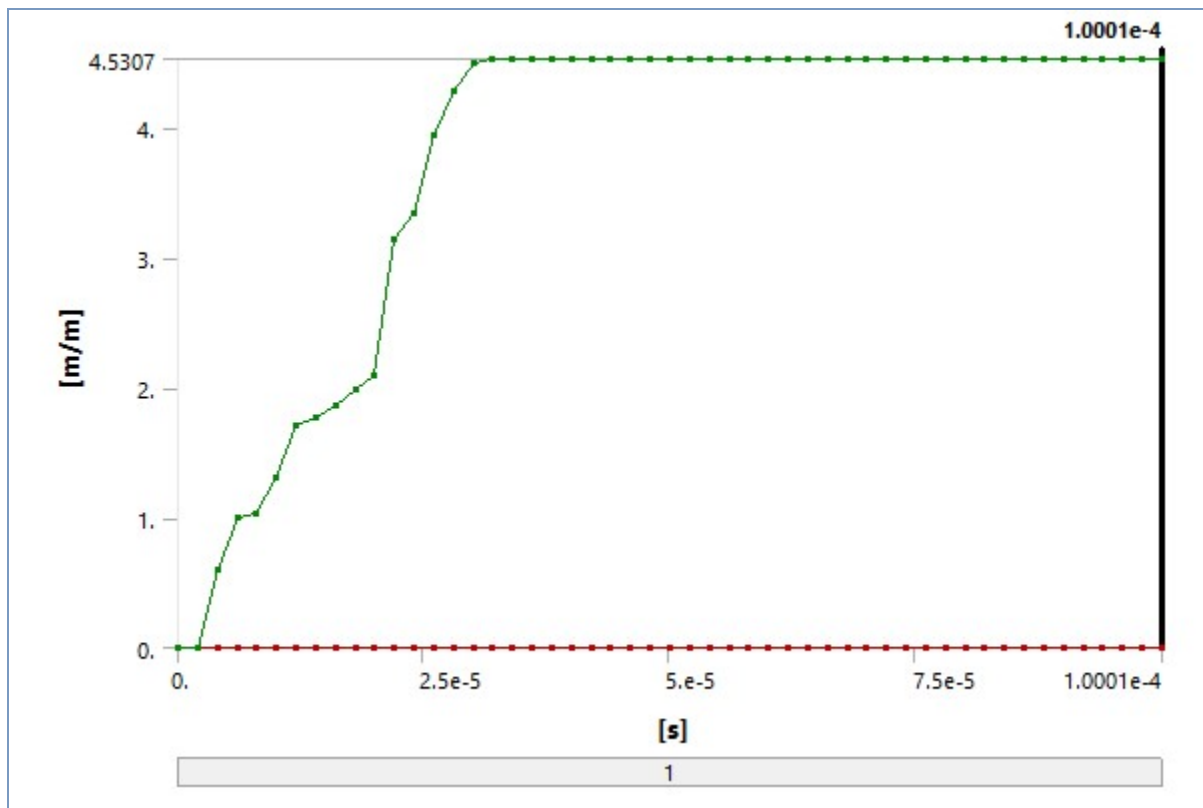


TABLE 21
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Equivalent Elastic Strain

Time [s]	Minimum [m/m]	Maximum [m/m]
1.1755e-038		0.
2.0034e-006		
4.0325e-006		0.59783
6.0106e-006		0.99758
8.0028e-006		1.0361
1.0012e-005		1.3171
1.2017e-005		1.7078
1.401e-005		1.7742
1.6003e-005		1.8587
1.8007e-005		1.9888
2.0001e-005		2.0927
2.2007e-005		3.1483
2.4004e-005		3.3376
2.6017e-005		3.9468
2.8007e-005		4.2788
3.0002e-005		4.4966
3.2003e-005		
3.4009e-005		
3.6003e-005		
3.8001e-005		
4.0003e-005		
4.2006e-005		
4.4006e-005		
4.6008e-005		
4.8009e-005		

5.0004e-005	0.	4.5307
5.2005e-005		
5.4011e-005		
5.601e-005		
5.8e-005		
6.0005e-005		
6.2005e-005		
6.4006e-005		
6.6001e-005		
6.8009e-005		
7.0009e-005		
7.2003e-005		
7.4005e-005		
7.6002e-005		
7.8004e-005		
8.0009e-005		
8.2001e-005		
8.4002e-005		
8.6e-005		
8.8002e-005		
9.0007e-005		
9.2002e-005		
9.4002e-005		
9.6e-005		
9.8007e-005		
1.0001e-004		

TABLE 22
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Probes

Object Name	<i>Velocity Probe</i>
State	Solved
Definition	
Type	Velocity
Location Method	Geometry Selection
Geometry	1 Face
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	Z Axis
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
Z Axis	11.937 m/s
Maximum Value Over Time	
Z Axis	750. m/s
Minimum Value Over Time	
Z Axis	7.3798 m/s
Information	
Time	1.0001e-004 s
Set	51
Cycle Number	8497

FIGURE 3

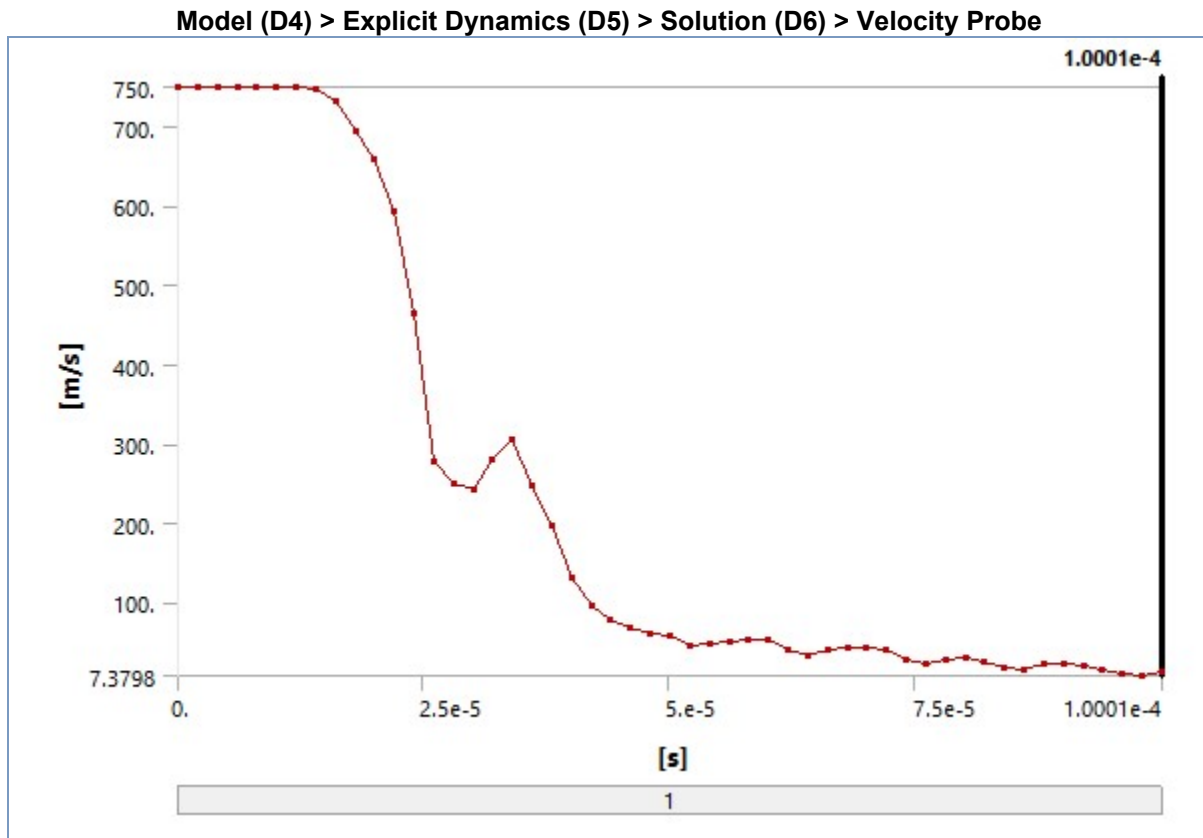


TABLE 23
Model (D4) > Explicit Dynamics (D5) > Solution (D6) > Velocity Probe

Time [s]	Velocity Probe (Z) [m/s]
1.1755e-038	750.
2.0034e-006	
4.0325e-006	
6.0106e-006	
8.0028e-006	
1.0012e-005	749.98
1.2017e-005	749.8
1.401e-005	747.86
1.6003e-005	732.19
1.8007e-005	693.2
2.0001e-005	660.07
2.2007e-005	593.12
2.4004e-005	465.77
2.6017e-005	277.5
2.8007e-005	250.94
3.0002e-005	243.2
3.2003e-005	279.98
3.4009e-005	305.08
3.6003e-005	248.43
3.8001e-005	196.06
4.0003e-005	130.
4.2006e-005	96.414
4.4006e-005	77.393
4.6008e-005	69.139

4.8009e-005	59.362
5.0004e-005	58.859
5.2005e-005	44.561
5.4011e-005	47.919
5.601e-005	50.771
5.8e-005	51.989
6.0005e-005	52.803
6.2005e-005	41.329
6.4006e-005	32.596
6.6001e-005	41.267
6.8009e-005	42.075
7.0009e-005	41.895
7.2003e-005	40.225
7.4005e-005	28.243
7.6002e-005	23.619
7.8004e-005	26.986
8.0009e-005	29.228
8.2001e-005	24.787
8.4002e-005	17.991
8.6e-005	15.666
8.8002e-005	22.141
9.0007e-005	21.474
9.2002e-005	20.284
9.4002e-005	14.25
9.6e-005	9.2458
9.8007e-005	7.3798
1.0001e-004	11.937

Material Data

armer grade

TABLE 24
armer grade > Constants

Density	7800 kg m ⁻³
Isotropic Secant Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat Constant Pressure	434 J kg ⁻¹ C ⁻¹
Isotropic Thermal Conductivity	60.5 W m ⁻¹ C ⁻¹
Isotropic Resistivity	1.7e-007 ohm m

TABLE 25
armer grade > Appearance

Red	Green	Blue
132	139	179

TABLE 26
armer grade > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0

TABLE 27
armer grade > Compressive Yield Strength

--

Compressive Yield Strength Pa
2.5e+008

TABLE 28
armer grade > Tensile Yield Strength

Tensile Yield Strength Pa
2.5e+008

TABLE 29
armer grade > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
4.6e+008

TABLE 30
armer grade > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 31
armer grade > Alternating Stress Mean Stress

Alternating Stress Pa	Cycles	Mean Stress Pa
3.999e+009	10	0
2.827e+009	20	0
1.896e+009	50	0
1.413e+009	100	0
1.069e+009	200	0
4.41e+008	2000	0
2.62e+008	10000	0
2.14e+008	20000	0
1.38e+008	1.e+005	0
1.14e+008	2.e+005	0
8.62e+007	1.e+006	0

TABLE 32
armer grade > Strain-Life Parameters

Strength Coefficient Pa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient Pa	Cyclic Strain Hardening Exponent
9.2e+008	-0.106	0.213	-0.47	1.e+009	0.2

TABLE 33
armer grade > Isotropic Elasticity

Temperature C	Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa
	2.1e+011	0.33	2.0588e+011	7.8947e+010

TABLE 34
armer grade > Isotropic Relative Permeability

Relative Permeability
10000

TABLE 35
armer grade > Johnson Cook Strength

Initial Yield Stress Pa	Hardening Constant Pa	Hardening Exponent	Strain Rate Constant	Thermal Softening Exponent	Melting Temperature C	Reference Strain Rate (/sec)

1.342e+009	1.5048e+008	0.2548	6.34e-002	0.8	1526.8	1
------------	-------------	--------	-----------	-----	--------	---

Lead Core

TABLE 36
Lead Core > Constants

Density	11300 kg m ⁻³
Isotropic Secant Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat Constant Pressure	129 J kg ⁻¹ C ⁻¹
Isotropic Thermal Conductivity	60.5 W m ⁻¹ C ⁻¹
Isotropic Resistivity	1.7e-007 ohm m

TABLE 37
Lead Core > Appearance

Red	Green	Blue
132	139	179

TABLE 38
Lead Core > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0

TABLE 39
Lead Core > Compressive Yield Strength

Compressive Yield Strength Pa
2.5e+008

TABLE 40
Lead Core > Tensile Yield Strength

Tensile Yield Strength Pa
2.5e+008

TABLE 41
Lead Core > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
4.6e+008

TABLE 42
Lead Core > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 43
Lead Core > Alternating Stress Mean Stress

Alternating Stress Pa	Cycles	Mean Stress Pa
3.999e+009	10	0
2.827e+009	20	0
1.896e+009	50	0
1.413e+009	100	0
1.069e+009	200	0
4.41e+008	2000	0
2.62e+008	10000	0
2.14e+008	20000	0

1.38e+008	1.e+005	0
1.14e+008	2.e+005	0
8.62e+007	1.e+006	0

TABLE 44
Lead Core > Strain-Life Parameters

Strength Coefficient Pa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient Pa	Cyclic Strain Hardening Exponent
9.2e+008	-0.106	0.213	-0.47	1.e+009	0.2

TABLE 45
Lead Core > Isotropic Elasticity

Temperature C	Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa
	1.3e+010	0.3	1.0833e+010	5.e+009

TABLE 46
Lead Core > Isotropic Relative Permeability

Relative Permeability
10000

TABLE 47
Lead Core > Johnson Cook Strength

Initial Yield Stress Pa	Hardening Constant Pa	Hardening Exponent	Strain Rate Constant	Thermal Softening Exponent	Melting Temperature C	Reference Strain Rate (/sec)
2.4e+007	3.e+008	1	0.1	1	486.85	1

Structural Steel

TABLE 48
Structural Steel > Constants

Density	8470 kg m ⁻³
Isotropic Secant Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat Constant Pressure	400 J kg ⁻¹ C ⁻¹
Isotropic Thermal Conductivity	60.5 W m ⁻¹ C ⁻¹
Isotropic Resistivity	1.7e-007 ohm m

TABLE 49
Structural Steel > Appearance

Red	Green	Blue
132	139	179

TABLE 50
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0

TABLE 51
Structural Steel > Compressive Yield Strength

Compressive Yield Strength Pa
2.5e+008

TABLE 52
Structural Steel > Tensile Yield Strength

Tensile Yield Strength Pa
2.5e+008

TABLE 53
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
4.6e+008

TABLE 54
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 55
Structural Steel > Alternating Stress Mean Stress

Alternating Stress Pa	Cycles	Mean Stress Pa
3.999e+009	10	0
2.827e+009	20	0
1.896e+009	50	0
1.413e+009	100	0
1.069e+009	200	0
4.41e+008	2000	0
2.62e+008	10000	0
2.14e+008	20000	0
1.38e+008	1.e+005	0
1.14e+008	2.e+005	0
8.62e+007	1.e+006	0

TABLE 56
Structural Steel > Strain-Life Parameters

Strength Coefficient Pa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient Pa	Cyclic Strain Hardening Exponent
9.2e+008	-0.106	0.213	-0.47	1.e+009	0.2

TABLE 57
Structural Steel > Isotropic Elasticity

Temperature C	Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa
	1.1e+011	0.34	1.1458e+011	4.1045e+010

TABLE 58
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000

TABLE 59
Structural Steel > Johnson Cook Strength

Initial Yield Stress Pa	Hardening Constant Pa	Hardening Exponent	Strain Rate Constant	Thermal Softening Exponent	Melting Temperature C	Reference Strain Rate (/sec)
1.2e+009	5.e+010	1	0	1	1526.8	1

Brass

TABLE 60

Brass > Constants

Density	8470 kg m ⁻³
Isotropic Secant Coefficient of Thermal Expansion	1.2e-005 C ⁻¹
Specific Heat Constant Pressure	400 J kg ⁻¹ C ⁻¹
Isotropic Thermal Conductivity	60.5 W m ⁻¹ C ⁻¹
Isotropic Resistivity	1.7e-007 ohm m

TABLE 61**Brass > Appearance**

Red	Green	Blue
132	139	179

TABLE 62**Brass > Compressive Ultimate Strength**

Compressive Ultimate Strength Pa
0

TABLE 63**Brass > Compressive Yield Strength**

Compressive Yield Strength Pa
2.5e+008

TABLE 64**Brass > Tensile Yield Strength**

Tensile Yield Strength Pa
2.5e+008

TABLE 65**Brass > Tensile Ultimate Strength**

Tensile Ultimate Strength Pa
4.6e+008

TABLE 66**Brass > Isotropic Secant Coefficient of Thermal Expansion**

Zero-Thermal-Strain Reference Temperature C
22

TABLE 67**Brass > Alternating Stress Mean Stress**

Alternating Stress Pa	Cycles	Mean Stress Pa
3.999e+009	10	0
2.827e+009	20	0
1.896e+009	50	0
1.413e+009	100	0
1.069e+009	200	0
4.41e+008	2000	0
2.62e+008	10000	0
2.14e+008	20000	0
1.38e+008	1.e+005	0
1.14e+008	2.e+005	0
8.62e+007	1.e+006	0

TABLE 68**Brass > Strain-Life Parameters**

Strength Coefficient Pa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient Pa	Cyclic Strain Hardening Exponent
9.2e+008	-0.106	0.213	-0.47	1.e+009	0.2

TABLE 69
Brass > Isotropic Elasticity

Temperature C	Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa
	1.1e+011	0.3	9.1667e+010	4.2308e+010

TABLE 70
Brass > Isotropic Relative Permeability

Relative Permeability
10000

TABLE 71
Brass > Johnson Cook Strength

Initial Yield Stress Pa	Hardening Constant Pa	Hardening Exponent	Strain Rate Constant	Thermal Softening Exponent	Melting Temperature C	Reference Strain Rate (/sec)
2.06e+008	5.05e+008	0.42	1.e-002	1.68	915.85	1